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Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory's Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes. While this parameter provides a useful baseline for morphological classification, its binary nature prevents the optimal combination of information from multiple bands, as well as from other classification methods (e.g., those based on colors or external datasets). In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\rm PSF} - m_{\rm CModel}$ and magnitude $m_{\rm CModel}$. At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which, with appropriately chosen priors for the star-to-galaxy number ratio, allows us to compute p_S, the posterior probability that a source is unresolved. We apply this method to the Rubin Data Preview 2 object catalog and validate the resulting classifications using external HST labels in the COSMOS field. Using these labels as the ground truth, we quantitatively compare the performance of the p_S-based classifier with that of the Rubin `extendedness` parameter. We find that the p_S-based classifier consistently outperforms the Rubin `extendedness` baseline, with the improvement increasing as additional bands are incorporated. Approximately, the p_S classifier achieves the same classification performance at a magnitude limit about half a magnitude fainter than the `extendedness` classifer. Depending on the scientific application, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

The morphological classification of resolved and unresolved sources[4], commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin's Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies and galaxies that exist are obviously extended on the sky. Star-galaxy separation becomes

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[4] Note that quasars and galaxies can be unresolved and even some stars can be resolved, such as Asymptotic Giant Branch stars.

increasingly harder at faint magnitudes, such as those probed by Rubin's Legacy Survey of Space and Time (LSST; for details see Ž. Ivezić et al. 2019a and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller. The selection of clean faint stellar samples is especially hard.

A variety of methods have been proposed in the literature for morphological classification of sources in images. C. T. Slater et al. (2020) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star-galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline (R. H. Lupton et al. 2002), or SExtractor pipeline (E. Bertin & S. Arnouts 1996), are closely related both to each other, and to the Bayesian model odds ratio, first derived by W. L. Sebok (1979). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best peformance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes, $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness=0`, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness=1`. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin's Data Preview 1 catalogs, see Figure 28 in Vera C. Rubin Observatory Team et al. (2026) or Figure 3 further below).

Although `extendedness` was successfuly used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star-galaxy separation threshold is optimized at the faint, low S/N, end of the data, the ability to recognize barely resolved objects at the bright end is not fully exploited. Furthermore, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands. With appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved), can be computed in Bayesian framwork.

The main goal of this paper is to implement the steps towards Bayesian star-galaxy separation outlined in SIL2020 using Rubin's Data Preview 2 data. We describe our dataset in §2, and in §3 discuss star-galaxy separation methodology, including quantification of its performance using HST-based labels. We discuss our results and how to retrieve our catalog with improved star-galaxy separation parameters in §4.

2. DATASETS

Here we describe two principal datasets used in our analysis: Rubin Data Preview 2 catalogs and HST data with star-galaxy labels used for testing methods discussed in §3.

2.1. *Rubin Data Preview 2*

All data collected with the LSST Camera Vera C. Rubin Observatory Team (2026) during the commissioning of the Rubin Observatory will be released as Data Preview 2[5] (hereafter DP2). Coadded deep images and resulting catalogs have been released in July 2026 as Early Data Preview 2. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky.

In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to only about 1 magnitude shallower limit than anticipated for 10-year LSST), and space-based HST COSMOS2020/FARMER star and galaxy labels are available (J. R. Weaver et al. 2022). As a control sample for testing code robustness, we also used Rubin data for the Extended Chandra Deep Field South (ECDFS). The Rubin $5-\sigma$ depth for point sources in the COSMOS field, estimated as the median magnitude of point sources with photometric errors of $\sim$0.2 mag, is (25.66, 26.66, 26.36, 25.91, 24.94, 23.59) in the *ugrizy* bands, respectively.

[5] See https://dp2.lsst.io/

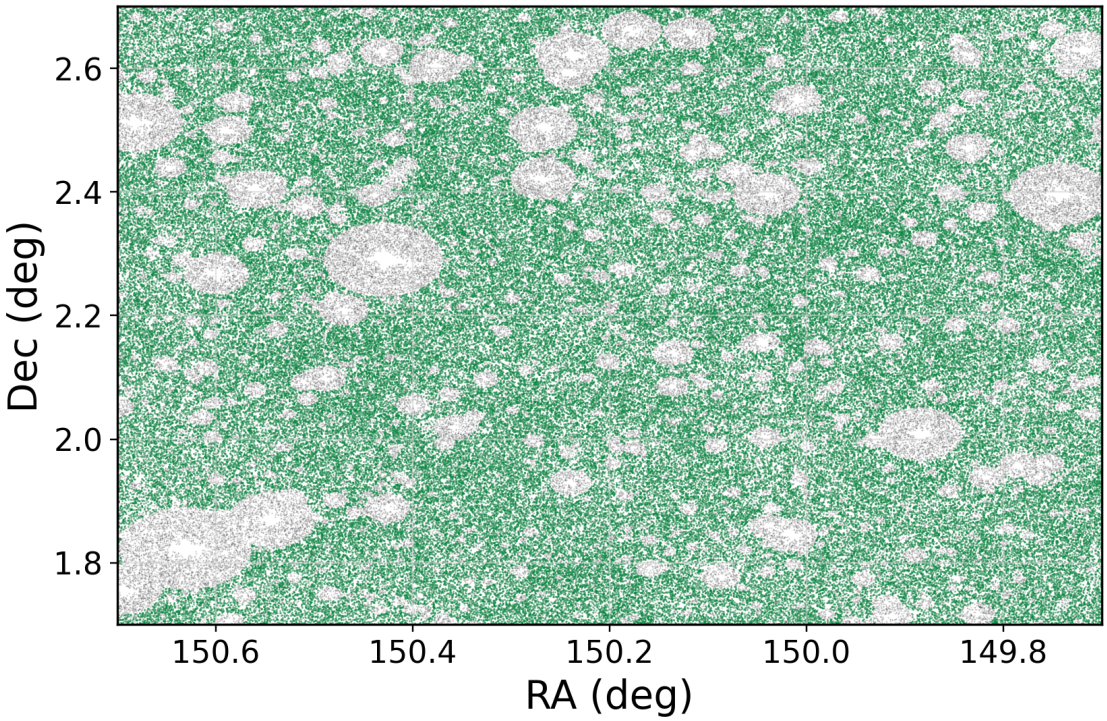


Figure 1. The footprint for the DP2-COSMOS sample analyzed here. Gray points show Rubin DP2 objects and green points show the subset matched to the HST COSMOS2020/FARMER sample a 0.5 arcsec matching radius. The white regions in the centers of gray areas correspond to areas where no objects are listed in either catalog.

2.2. *HST Star-galaxy Classification Labels in the COSMOS Field*

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \mathrm{RA} < 150.7$ and $1.7 < \mathrm{Dec} < 2.7$, with $16 \leq r_{\mathrm{CModel}} < 26$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER catalog (see Figure 1).

The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using HST's `truth_binary` = 1 for stars and `truth_binary` = 0 for galaxies. Their number counts as a function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit $r \sim 26$) and high Galactic latitude ($b \sim 50°$).

2.3. *The color-magnitude distributions of sources classified using HST labels*

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is likely that some very faint sources classified by HST as stars might be unresolved galaxies even at HST's resolution. Therefore, the HST labels will help us assess whether a source should be resolved or unresolved in Rubin images, but do not necessarily provide definitive information about astrophysical nature of a source (e.g., to separate unresolved stars and galaxies, one would need spectra or broad-band photometry over a much wider weavelength range, say, in near-infrared K band). Furthermore, no classification is perfect and thus we first examine the distribution of HST-classified "stars" and "galaxies" in various informative color-color diagrams constructed with Rubin photometry.

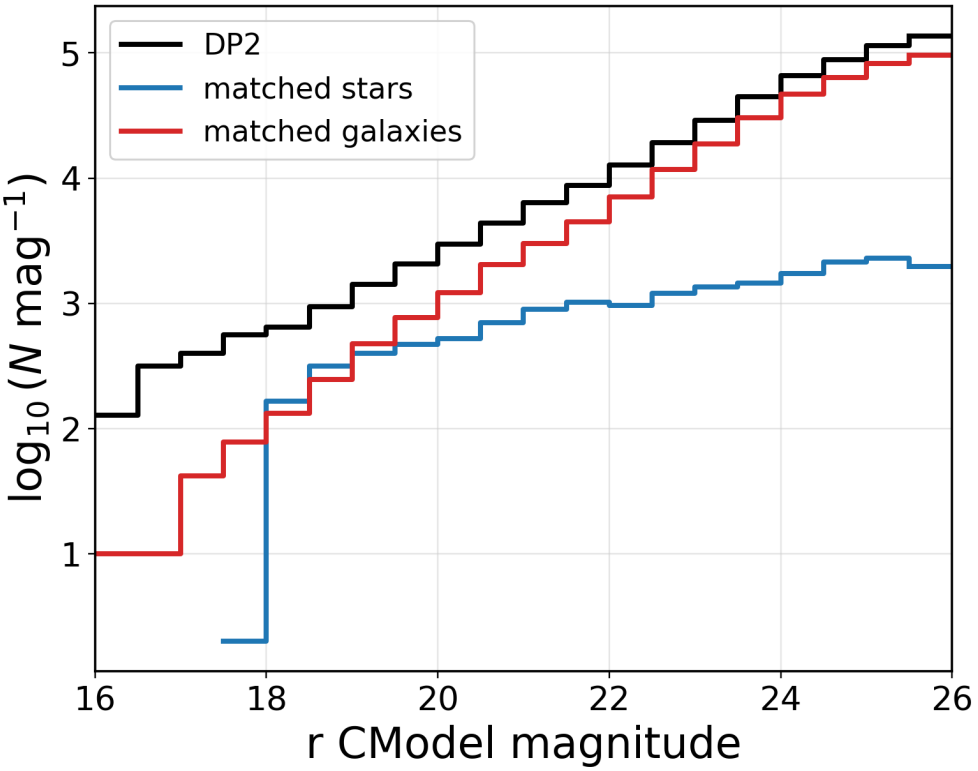


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins. Note that galaxies outnumber stars at the faint end by more than an order of magnitude.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\rm PSF} - m_{\rm CModel}$ vs. $m_{\rm CModel}$ diagram. As we describe in detail in the next Section, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method presented here.

The top right panel in Figure 3 shows that faint galaxies ($r > 24$), in addition to outnumbering stars at faint magnitudes, have on average bluer $g - i$ colors than stars. The clearly different distributions of stars and galaxies in color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 clearly show that increased photometric noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. Extending the start of this deterioration to a fainter magnitude limit is one of the primary goals of this work. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u - g$ color more typical of low-redshift quasars than of main sequence stars (for a detailed discussion of the blue star counts see L. Palaversa et al. 2026).

2.4. *The color-magnitude distributions of sources classified using* `extendedness` *parameter*

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin `extendedness` binary class in the r band. As evident, a significant fraction of faint sources within about two magnitudes from the faint limit are misclassified when using default `extendedness` parameter. Misclassifications have a more severe impact on stellar sample: only 26% of "stars" selected by `extendedness`=0 condition are unresolved in HST data. This high level of stellar sample contamination by faint galaxies increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for "stars" (objects unresolved by HST) is only 67%. Due to much higher counts at faint magnitudes and high Galactic latitudes,

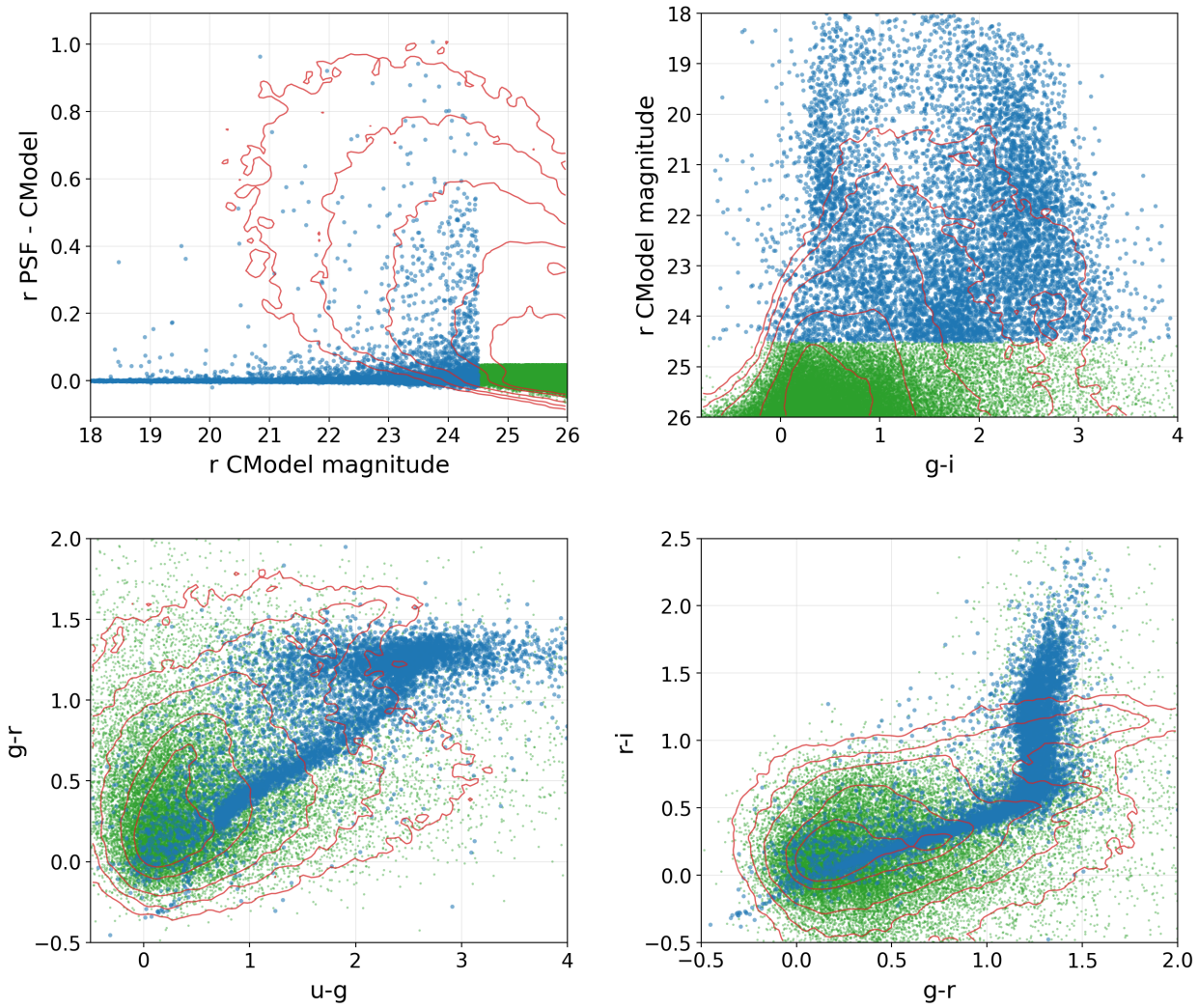


Figure 3. The distribution of stars with r-band CModel magnitude < 24.5(blue dots) and galaxies (red countours), classified using HST labels, in various infomative diagrams. Green points show Rubin DP2 sources classified as unresolved (extendedness$_r = 0$) with r-band CModel magnitude $24.5 < r < 26$ and $m_{\rm PSF} - m_{\rm CModel} < 0.05$. The CModel-magnitude based colors are corrected for the ISM dust extinction, while the r-band CModel magnitude is not. The top left panel shows the $\Delta m = m_{\rm PSF} - m_{\rm CModel}$ vs. $m_{\rm CModel}$ diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and for galaxies $\Delta m > 0$. The top right panel shows the r vs. $g - i$ color magnitude diagram and two bottom panels show most informative color-color diagrams. The main stellar locus, dominated by main sequence stars and red giants, is clearly visible with its blue end redwards from $u - g \sim 0.6$; unresolved sources with $u - g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries. In the region with $24 < r < 24.5$ and $0.0 < g - i < 1.0$, galaxies outnumber stars by a factor of 80.

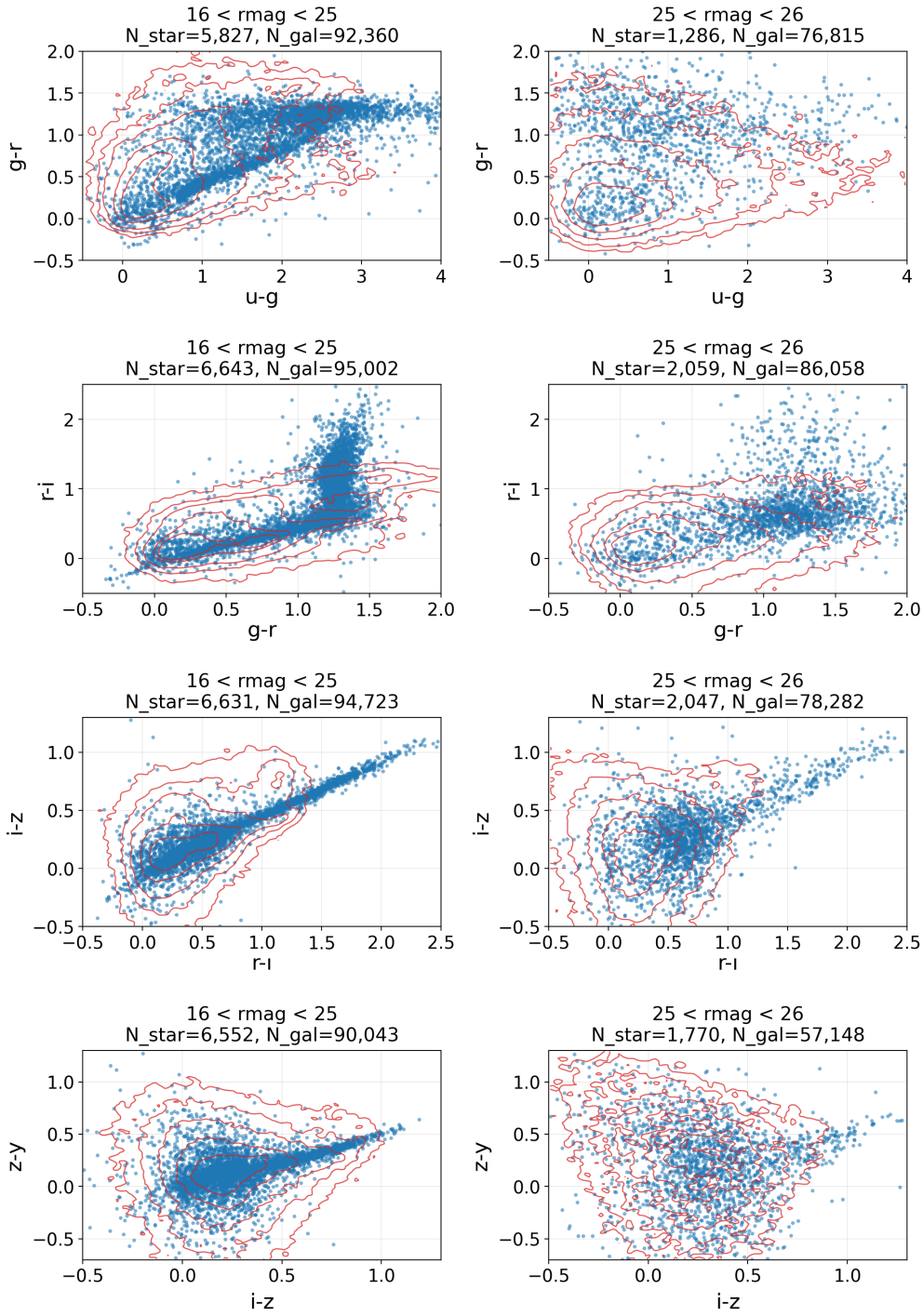


Figure 4. Similar to the bottom two panels in Figure 3, except that here all principal color-color diagrams are shown and the data are split by the r-band CModel magnitude to show deterioration of photometry close to the faint magnitude limit (left: 16 < rmag < 25; right: 25 < rmag < 26). The $5-\sigma$ depth for point sources in the r band for this dataset is 26.36.

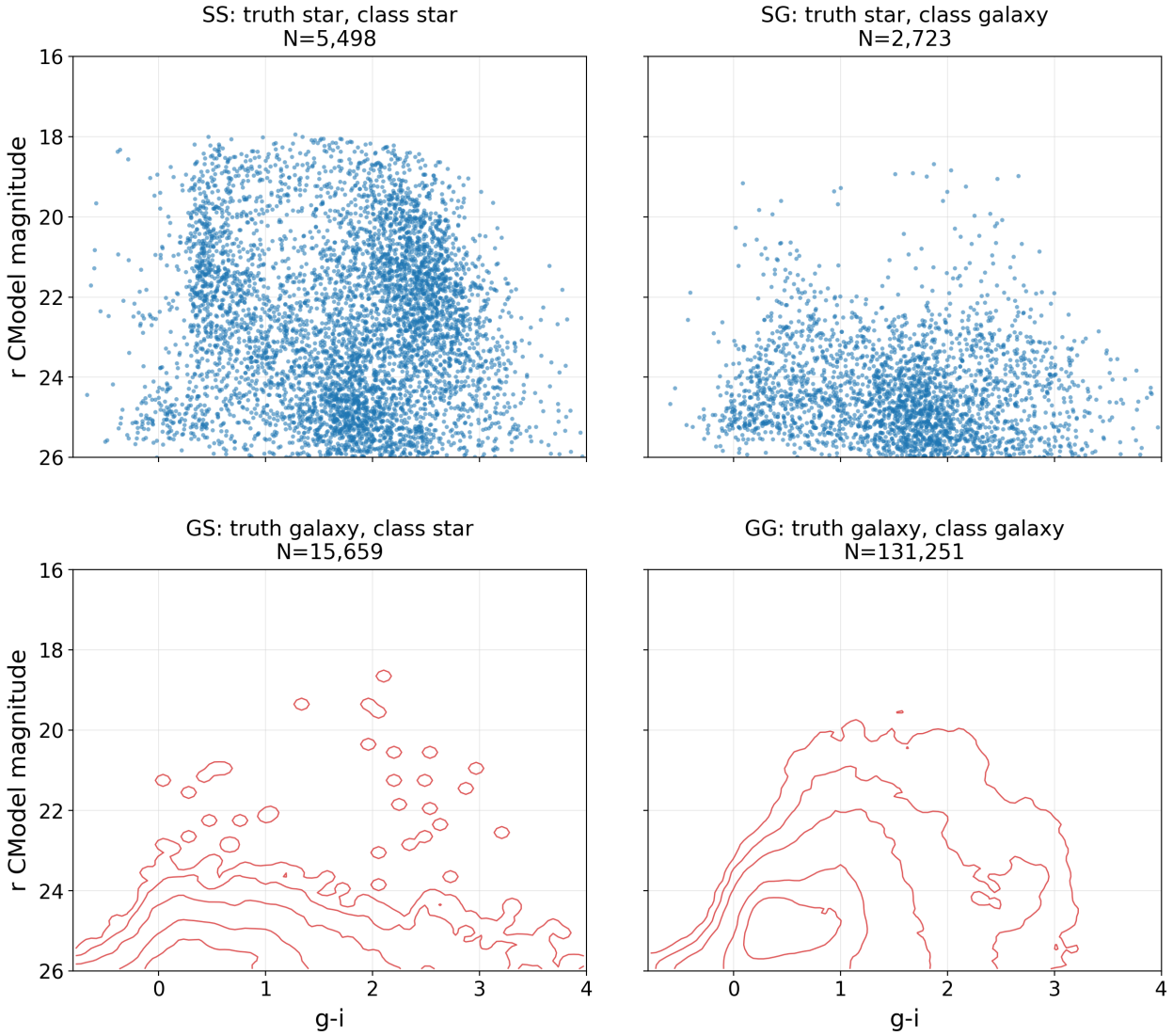


Figure 5. Performance of Rubin's default `extendedness` parameter in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g-i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by `extendedness`=0 is severely contaminated by galaxies for $r_{\rm CModel} > 24$ (bottom left panel). In the region with $24 < r < 24.5$ and $0.0 < g-i < 1.0$, misclassied galaxies (bottom left panel) outnumber selected stars (top left panel) by a factor of 5.5.

the purity and completeness for galaxies are much higher: 98% and 89%, respectively. In other words, a simple classifier that declares every object to be a galaxy would formally perform quite well. Therefore, we will focus our analysis in the next Section on the classifier's performance when selecting unresolved objects.

2.5. *Suspect HST Classification Labels*

There is a subset of objects classified as "stars" (unresolved) by HST which appear clearly resolved when using Rubin's default `extendedness` parameter. Figure 6 shows their distribution in color-color diagrams, with symbols color-coded by Δm in the r band. As evident, their color-color distribution is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars (see Figures 2 and 4 in V. Smolčić et al. 2004.) To test this hypothesis, we visually inspected several hundred images (443 objects with HST label "star", $r < 23$ and $\Delta m > 0.05$ mag; see Figure 7 for an example). Only 1.4% of selected objects are detected as "isolated" (computed using processing flag "detect_isIsolated") and about

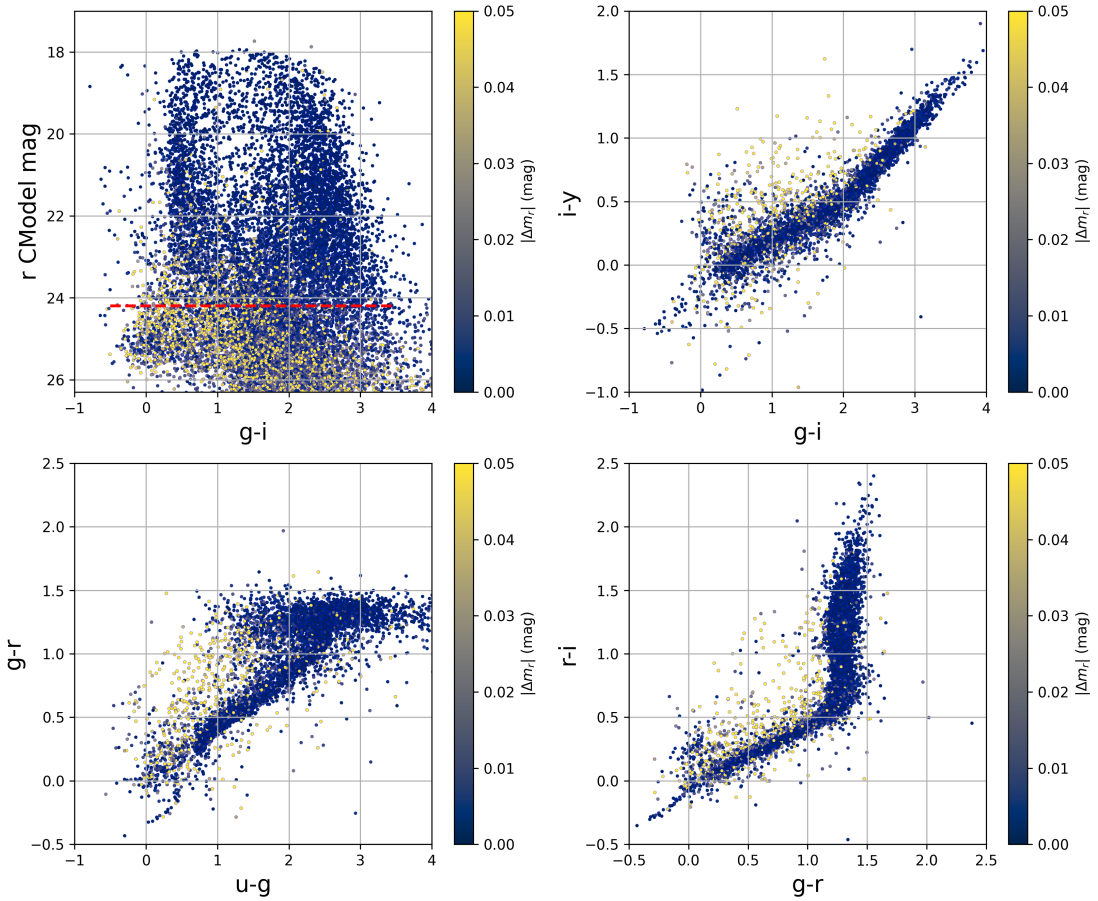


Figure 6. The distribution of objects classified as "stars" by HST, but which appear clearly resolved when using Rubin's default `extendedness` parameter, in color-magnitude (top left) and informative color-color diagrams. Only objects with $r < 24.2$ are shown in the three color-color diagrams. Symbols are color-coded by the absolute value of Rubin's Δm in the r band. The color-color distribution of objects with $|\Delta m| > 0.05$ mag (about 3% of all objects classified by HST as "stars" and brighter than $r = 23$) is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars.

half seem to have a nearby stellar-like companion. In some cases a bright source or an extended galaxy is nearby, and about one tenth or fewer appear bluish and consistent with quasars. Detailed examination of such cases is beyond the scope of this work.

Because we aim to test statistical performance of our star-galaxy separation method on unambiguous objects, we exclude objects with HST "star" label that both appear resolved in Rubin data and also have Rubin colors inconsistent with the main stellar locus.

We checked whether HST-labeled stars lie near the empirical COSMOS stellar locus. For each source, we calculated its shortest distance to the locus using five dereddened CModel colors: ($u-g$, $g-r$, $r-i$, $i-z$, $z-y$). Because the scatter in these colors changes with magnitude, we set a separate distance threshold in each Rubin r-band CModel magnitude bin. Each threshold was set to include at least 80% of HST-labeled stars with measurements in all five colors.

We excluded an HST-labeled star from the performance sample only when it was both beyond the color threshold and classified as resolved by Rubin in the r band (extendedness$_r = 1$). This removed 836 of the 15,760 HST-labeled stars (5.30%), leaving 14,924 (94.70%). In the two magnitude ranges of interest, the exclusions were 104 of 1,388 stars (7.49%) at $24 < r < 24.5$ and 116 of 1,700 stars (6.82%) at $24.5 < r < 25$. Stars without the measurements needed to check both criteria were not excluded by this procedure.

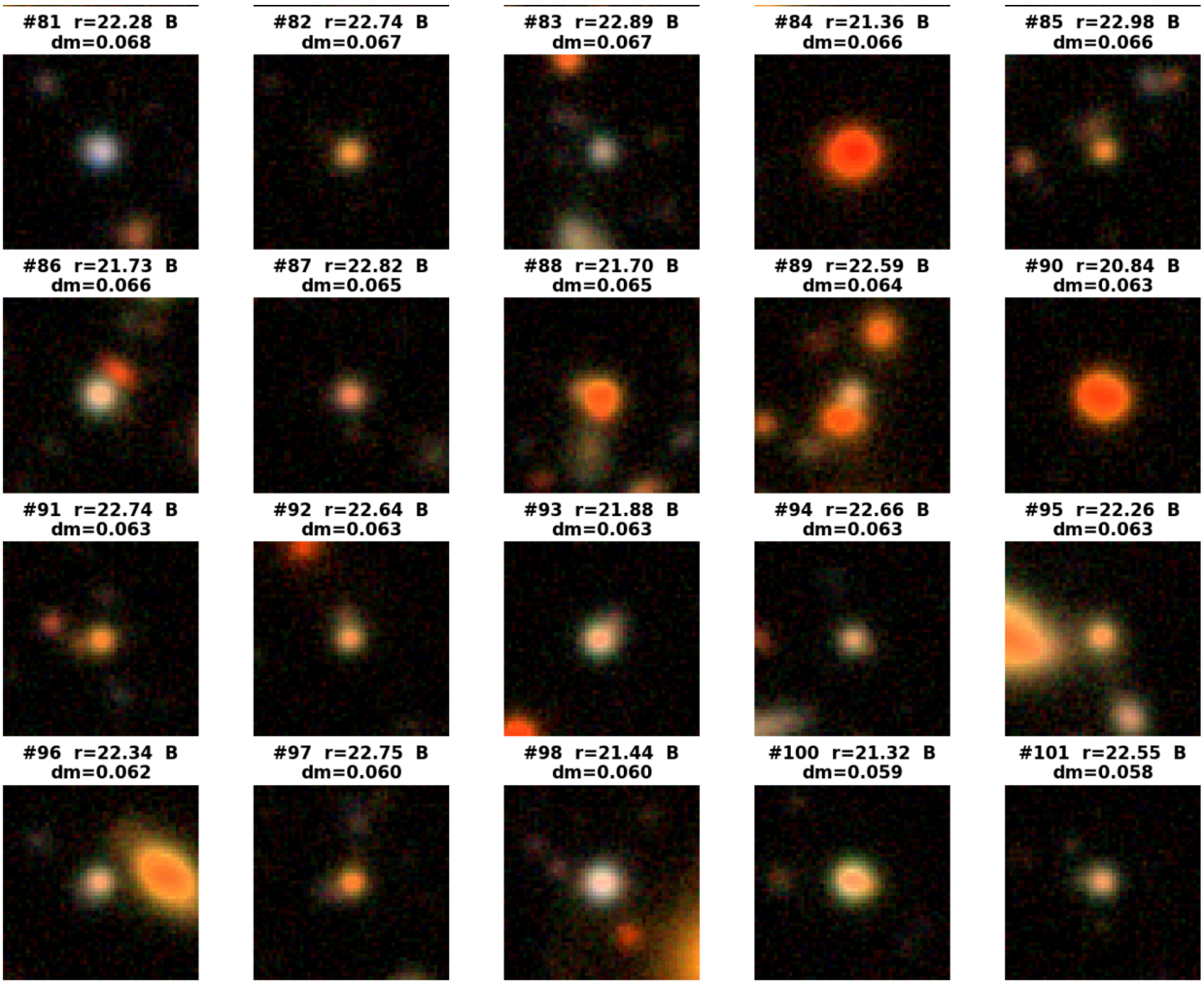


Figure 7. The Rubin *gri* color composite images extracted from "deep_coadd", 10 arcsec by 10 arcsec (50x50 pixels), for a random subsample of 443 objects brighter than $r = 23$ and selected as "stars" in HST and resolved in Rubin. About 98.6% are detected as blended objects.

3. STAR-GALAXY SEPARATION METHODOLOGY

We first briefly review the Bayesian star-galaxy separation method proposed in SIL2020 and then apply it to single-band Rubin data, followed by its extension to the multi-band case. Our principal measured object property is the difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes, $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is consistent with zero within the measurement noise, while for resolved objects $\Delta m > 0$ (see the top left panel in Figure 3). A detailed comparison and statistical discussion of this quantity and several other quantities proposed for star-galaxy separation in the literature is available in SIL2020.

3.1. *Bayesian Star-Galaxy Separation Method from SIL2020*

We first focus on the single-band case and then extend our discussion to the multi-band case. Using realistic simulations, SIL2020 showed that adopting a fixed, predefined threshold for Δm to select resolved objects is not optimal. The threshold should depend on magnitude because of variations in the object populations and signal-to-noise ratio due to statistical noise, as well as on sky location because the galaxy-to-star count ratio varies across the sky. Moreover, for an optimal combination of information from different bands or from other classification methods (e.g., based on colors), the observed Δm distribution needs to be transformed into a proper probability density function.

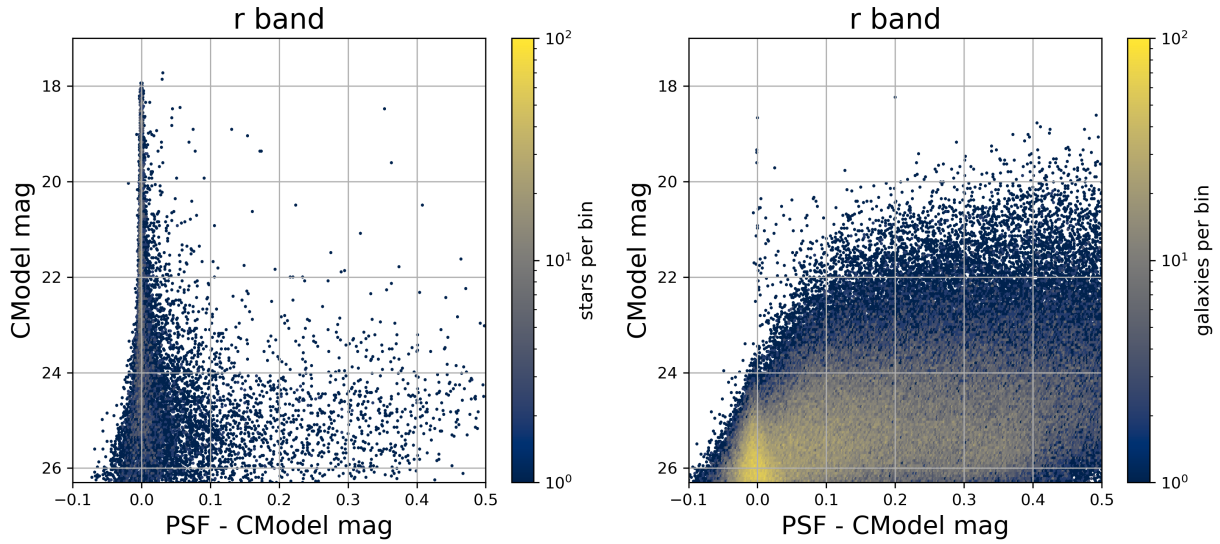


Figure 8. The panels show the Δm vs. $m_{\rm CModel}$ distributions for unresolved (left) and resolved (right) objects classified using HST labels. Note that some objects in the left panel appear clearly resolved by Rubin ($\Delta{\rm m} > 0.05$), even for $r < 24$ (see §2.5), and that galaxies (right panel) dominate the region around $\Delta{\rm m} \sim 0$ at the faint end ($r > 24$). Quantitative modeling of these two-dimensional distributions is at the core of the star-galaxy separation method discussed here.

The subsample of objects with HST labels can be used to quantify the Δm distribution separately for unresolved and resolved objects (see Figure 8). However, such a "training" sample is currently available only for a very small fraction of the LSST footprint (less than 1% of the area). Instead, we use the fact that the very large numbers of objects remaining even after binning in both sky location and magnitude can robustly constrain the two components of the Δm distribution even when such labels are not available a priori. We only use the subsample of objects with HST labels to independently assess the performance of our method.

3.2. *Mixture Model for Δm vs. $m_{\rm CModel}$ diagram*

We assume that a sample is selected from a suitably chosen sky region over which the stellar and galaxy populations can be assumed to be statistically uniform. Over angular scales of several degrees, stellar counts vary much more rapidly than galaxy counts (large galaxy clusters are an exception, but not important for the subsequent discussion). Based on the analysis of state-of-the-art Milky Way stellar population models by L. Palaversa et al. (2025), a sky region of several square degrees, corresponding to Rubin tracts[6] is sufficiently small for this purpose (and consistent with our choice of the COSMOS field for quantitative analysis).

Given a "tract" catalog, at a fixed CModel magnitude (see below for binning details), the distribution of Δm in a given band (with the band index suppressed for notational simplicity) is modeled as a mixture of an unresolved (star) component and a resolved (galaxy) component,

$$p(\Delta{\rm m} \,|\, m) = w_U \, f_{\rm unr}(\Delta{\rm m}) + w_R \, f_{\rm res}(\Delta{\rm m}), \tag{1}$$

where $w_U + w_R = 1$, and $f_{\rm unr}$ and $f_{\rm res}$ are properly normalized probability density functions. With this decomposition, the probability that an object is unresolved is given by Bayes' theorem as

$$p_S(\Delta{\rm m}) = p({\rm unr} \mid \Delta{\rm m}) = \frac{w_U f_{\rm unr}(\Delta{\rm m})}{w_U f_{\rm unr}(\Delta{\rm m}) + w_R f_{\rm res}(\Delta{\rm m})}. \tag{2}$$

[6] In the Rubin Science Pipelines, tracts and patches are the two levels of spatial subdivision used for coadded images and object catalogs. They are designed to make processing and storage scalable while preserving continuity across the sky. A tract is a relatively large region of the sky, approximately 1.75° × 1.75° (the exact size depends on the sky map projection). that is processed as a unit. Tracts overlap slightly with neighboring tracts to avoid edge effects. Each tract is subdivided into a regular grid of patches, with each tract in DP2 containing 81 patches. The patch size is about 12 arcmin × 12 arcmin (approximately, one CCD).

and the probability that it is resolved is $p_G = 1 - p_S$ (note that we use indices S and unr, or G and res interchangeably).

The same model predicts the performance, completeness, and contamination for each class as a function of a hard Δm threshold. If $\Delta\mathrm{m} < \Delta\mathrm{m}_{\mathrm{cut}}$ is classified as unresolved (S),

$$\mathrm{Completeness}(S) = F_{\mathrm{unr}}(\Delta\mathrm{m}_{\mathrm{cut}}) = \int_{-\infty}^{\Delta\mathrm{m}_{\mathrm{cut}}} f_{\mathrm{unr}}(\Delta\mathrm{m}) d\Delta\mathrm{m}, \tag{3}$$

and

$$\mathrm{Contamination}(S) = \frac{w_R F_{\mathrm{res}}(\Delta\mathrm{m}_{\mathrm{cut}})}{w_U F_{\mathrm{unr}}(\Delta\mathrm{m}_{\mathrm{cut}}) + w_R F_{\mathrm{res}}(\Delta\mathrm{m}_{\mathrm{cut}})}. \tag{4}$$

We test these predictions below using HST-based labels.

Therefore, a good model for $p(\Delta\mathrm{m}\,|\,m)$ (eq. 1) is crucial. Informed by observed Δm distributions (see Figure 8), we model the unresolved component, f_{unr}, as a single Gaussian with mean μ_U, width σ_U (and weight w_U). The resolved component is modeled as a sum of K log-normal distributions with weight $w_{R,k}$, location $\mu_{R,k}$, scale $\sigma_{R,k}$, and shape parameter $\alpha_{R,k}$. The full mixture probability density function is

$$p(\Delta m\,|\,m) = w_U\,\mathcal{N}(\Delta m|\,\mu_U,\,\sigma_U) + \sum_{k=1}^{K} w_{R,k}\,\mathcal{LN}(\Delta m|\,\mu_{R,k},\,\sigma_{R,k},\,\alpha_{R,k}), \tag{5}$$

where $\mathcal{N}$ denotes the Gaussian and $\mathcal{LN}$ the log-normal distribution, and the weights are constrained to sum to unity: $w_U = 1 - \sum_k w_{R,k}$ (functions $\mathcal{N}$ and $\mathcal{LN}$ are properly normalized probability density functions). The galaxy-to-star count ratio, which serves as Bayesian prior, is

$$N_{GS} \equiv \frac{N_G}{N_S} = \frac{\sum_{k=1}^{K} w_{R,k}}{w_U}. \tag{6}$$

The expression in eq. 2 can be recast as

$$p_S(\Delta\mathrm{m}) = \frac{1}{1 + N_{GS}\,BF}, \tag{7}$$

where the Bayes factor (the likelihood ratio for the galaxy and star hypotheses) is

$$BF = \frac{f_{\mathrm{res}}}{f_{\mathrm{unr}}}. \tag{8}$$

With $N_{GS} \sim 30$ (see Figure 9), it is sufficient to have $BF \sim 0.033$ for p_S to drop below 0.5 (because the sample is highly imbalanced). This decomposition in terms of prior and Bayes factor will be important later when combining multiple sources of classification information.

3.2.1. *Fitting Details*

The fitting is performed independently in bins of CModel magnitude. At bright magnitudes ($m_{\mathrm{CModel}} < 22$), where the resolved population is relatively simple, we use $K = 2$ resolved components in bins of width 1–3 mag spanning $15 < m_{\mathrm{CModel}} < 22$. At faint magnitudes ($m_{\mathrm{CModel}} \geq 22$), where the galaxy morphology distribution is more complex, we use $K = 3$ resolved components in finer bins of width 0.5–1 mag spanning $22 \leq m_{\mathrm{CModel}} < 27$. Within each magnitude bin, Δm values in the range $(-0.5, 1.5)$ are binned into 300 bins and normalized to unity. The mixture model is fit to these histograms using bounded least-squares optimization with Poisson-like uncertainties $\sigma_i = \sqrt{N_i + 1}\,/\,(N\,\delta)$, where N_i is the count in bin i, N is the total number of sources in the magnitude slice, and δ is the histogram bin width.

This fitting procedure is carried out in three steps. In the first step, all mixture parameters are fit independently in each magnitude bin. This free fit works well at bright magnitudes, where unresolved and resolved sources are clearly separated in Δm. However, at faint magnitudes ($m_{\mathrm{CModel}} \gtrsim 24$), photometric noise broadens the unresolved Gaussian until it becomes degenerate with the resolved components. The

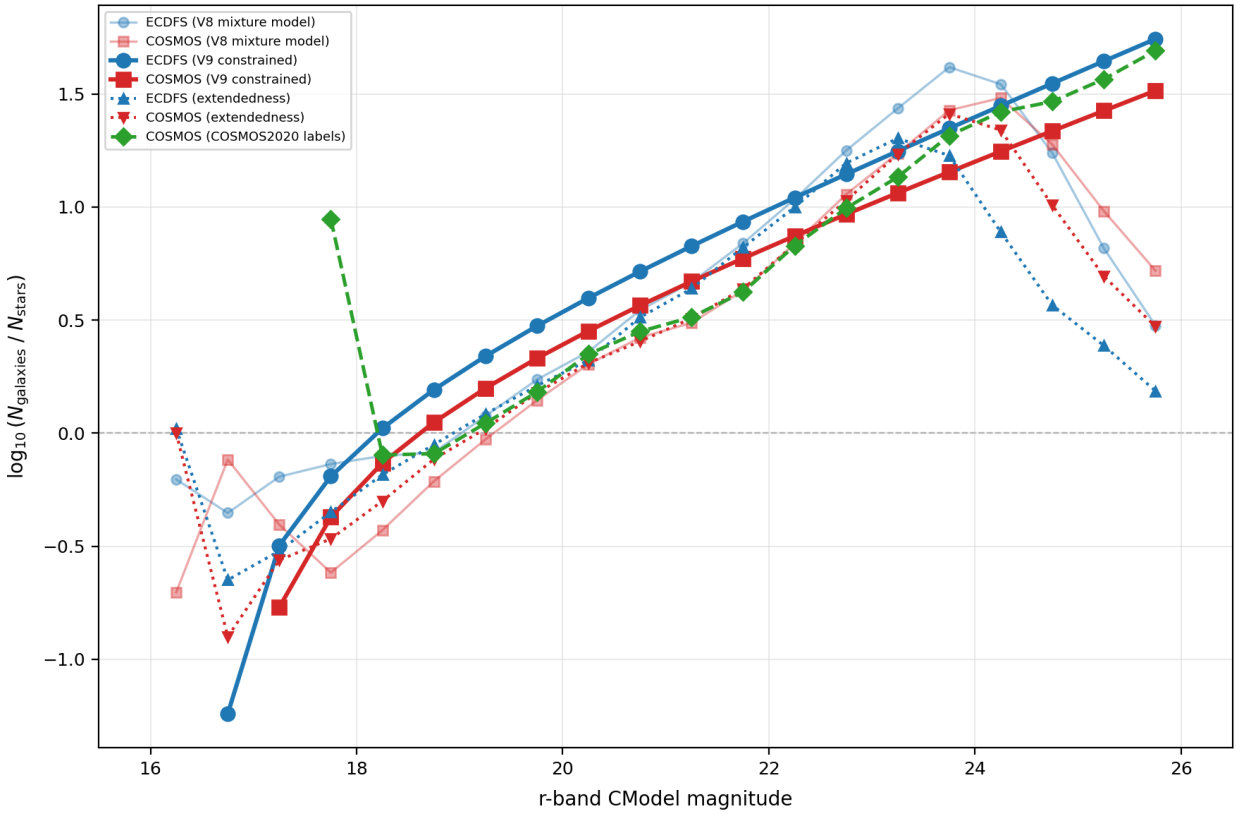


Figure 9. Galaxy-to-star count ratio $\log_{10}(N_{\rm gal}/N_{\rm star})$ vs. r-band CModel magnitude. The unconstrained mixture model (marked V8, faded curves) exhibits a spurious turnover at $m \gtrsim 24$ due to the unresolved Gaussian absorbing galaxy flux. The constrained model (marked V9, bold curves) recovers the expected monotonically increasing ratio, in good agreement with the COSMOS2020 truth labels (green dashed). Dotted lines show the Rubin `extendedness`-based classification for comparison (which also shows spurious turnover at the faint end).

fitter can no longer reliably distinguish the two populations, causing the unresolved Gaussian to absorb flux from the galaxy distribution. This inflates the inferred star fraction at faint magnitudes.

To prevent this degeneracy, we impose smooth constraints on the unresolved component. In the second step, the per-bin star parameters from step 1 are replaced with smooth functions of magnitude: μ_U is fixed to a single value (the median across all bins, typically ≈ 0 in each band); $\sigma_U(m)$ is fit with the photometric error model from B. Sesar et al. (2007) (their eq. 5),

$$\sigma_U(m) = \sqrt{\sigma_0^2 + a^2 \cdot 10^{\,b\,(m-25)}}, \tag{9}$$

where σ_0, a, and b are free parameters; and $w_U(m)$ is modeled as a linear fit to w_U versus magnitude, using only bins brighter than $m_{\rm CModel} = 24$ where the free fit is reliable. This smooth model is then extrapolated to fainter magnitude bins. In the third step, with the unresolved component (μ_U, σ_U, w_U) fixed to the smooth curves from step 2, only the resolved log-normal parameters are re-fit in each magnitude bin. The resolved weights are rescaled to sum to $(1 - w_U)$ so that the total mixture remains a proper probability density. The effectiveness of this constrained fitting procedure is demonstrated in Figure 9, which compares the galaxy-to-star count ratio $N_{\rm gal}/N_{\rm star}$ as a function of magnitude for models with and without the smooth constraints. The unconstrained model exhibits a spurious turnover at faint magnitudes, while the constrained model recovers the expected monotonically increasing ratio in agreement with the COSMOS2020 external truth labels (note that in this sample at the faint end galaxies outnumber stars by more than a factor of 30).

The model parameters are selected according to the CModel magnitude bin in which each source falls. The quality of the fits is illustrated in Figure 10, which shows the mixture model overlaid on the observed

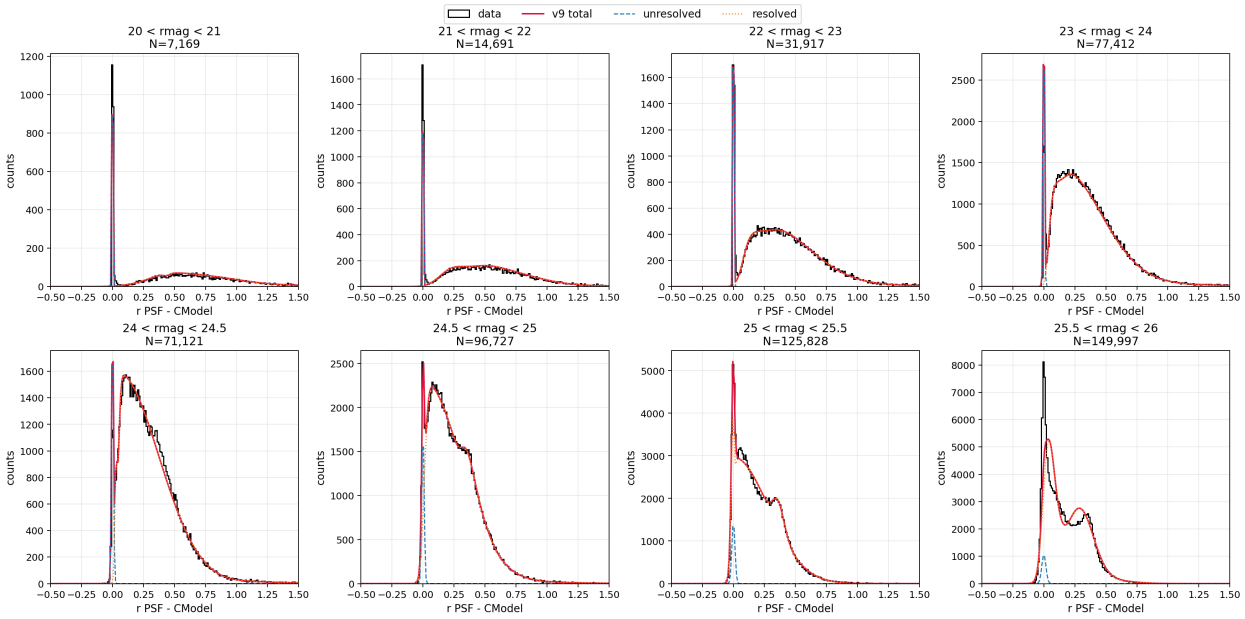


Figure 10. Observed Δm histograms (black) and fitted mixture models (red) for the r-band in each CModel magnitude bin. The unresolved Gaussian component is shown as a blue dashed line and the resolved log-normal components as orange dotted lines.

Δm histograms in each magnitude bin, and in Figure 11, which shows the 2D data, model, and residual maps in the Δm vs. $m_{\rm CModel}$ plane.

3.3. *The Performance of the Δm Mixture Model*

Here we use HST labels to evaluate completeness and contamination for the stellar sample and compare them with the mixture-model predictions. We focus on the stellar subsample and the "transition" magnitude range, $24 < r < 24.5$. Brighter than this range, population mixing is much smaller, leading to more accurate star-galaxy separation, whereas at fainter magnitudes the performance deteriorates rapidly (especially for the stellar subsample).

3.4. *The Multi-band Case*

We expect classification to improve when Δm measurements from multiple bands are used. However, combining information from different bands is a nontrivial statistical problem that is made more complex by astrophysical considerations. For example, galaxy morphology is wavelength dependent: different bands emphasize different stellar populations, dust distributions, star-forming regions, bulges, and disks, and thus a galaxy can show strong resolved structure in one band while being close to a PSF in another. A decisive measurement in one band should therefore not necessarily be diluted by several bands with weak or absent evidence. Consequently, combining measurements from multiple bands is not as straightforward as combining multiple measurements from a single band using eq. 7 (see Section 5 in SIL2020 for details).

An optimal Bayesian approach would be to generalize eq. 1 to a vector form and model the true object class as an underlying latent variable. The multi-band measurements are correlated projections of this latent structure rather than independent measurements of one scalar size. We explored a Gaussian copula construction for combining single-band probabilities, but it did not provide a satisfactory description of the desired behavior. We therefore adopt a different approach: first combine the measured Δm values across bands into a single statistic and then derive the final probabilities from its distribution, analogously to the single-band case.

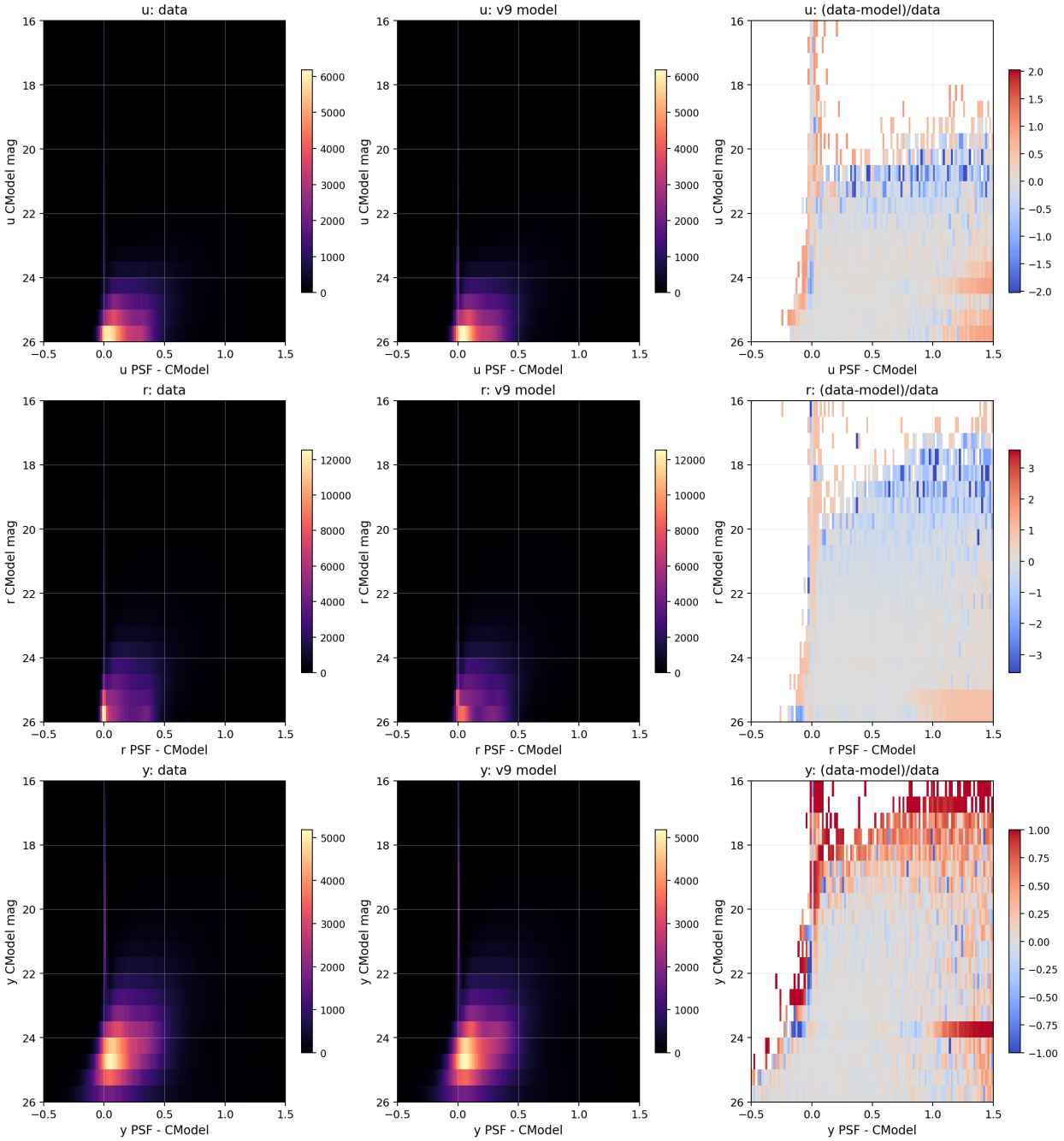


Figure 11. Two-dimensional comparison of observed source counts (left), V9 mixture-model counts (center), and fractional residuals (right) in the Δm vs. m_{CModel} plane, for the u, r, and y bands from top to bottom. The first two columns show counts without a square-root transformation and share the same linear color scale within each row. The right column shows $(N_{\mathrm{data}} - N_{\mathrm{model}})/N_{\mathrm{data}}$, with cells containing $N_{\mathrm{data}} = 0$ masked. Residual color limits are symmetric about zero and set separately for each band, as shown by the individual colorbars.

Motivated by the fact that the multi-band Δm distribution for the unresolved component can be described as a multivariate Gaussian with diagonal covariance matrix, we define a standardized variable in each band

$$D_i = \frac{\Delta m_i - \mu_{U,i}}{\sigma_{U,i}}, \tag{10}$$

where i is the band index and μ_U and σ_U are the best-fit parameter values for the Gaussian fit to the unresolved component (recall that $\mu_U \sim 0$). The bands may be all of *ugrizy* or any subset of N usable bands, B. We then define a radial metric

$$D = \left(\sum_B D_i^2 \right)^{1/2}. \tag{11}$$

When $N = 1$, $D \sim |\Delta m/\sigma_U|$, and it measures the distance from the origin in units of σ_U. For the unresolved component, D follows a χ-distribution with one degree of freedom, χ_1, which is also the half-normal distribution. If μ_U cannot be ignored, D follows a noncentral χ distribution. When $N > 1$, for the unresolved component $D \sim \chi_N$, with an expectation value approximately equal to $\langle D \rangle = \sqrt{N - 1/2}$.

Depending on the desired completeness for the unresolved component, a hard D threshold of several times $\sqrt{N}$ can be used for sample classification (this is a standard case of hypothesis testing in frequentist statistics). However, without a quantitative description of the resolved component as well, the resulting sample contamination cannot be estimated.

Figure 12 shows the overlap between the HST stars and HST galaxies D distributions and its dependence on the magnitude and band combination.

Even if the single-band Δm distribution for the resolved component was well described by a single log-normal distribution in every band, the resulting D distribution for the resolved component would be a complex function because of strong band-to-band correlations (for a toy model, see Appendix A). For this reason, we again use a linear combination of log-normal distributions to describe the D distribution for the resolved component. To provide more flexibility than χ_N, for the unresolved component we use the function

$$g(D) = C\, D^a\, \exp(-b\, D^2), \tag{12}$$

where C, a, and b are free parameters. Note that for the χ_N distribution, $a = N - 1$ and $b = 1/2$.

In summary, given the best-fit decomposition (analogous to eq. 1)

$$p(D \,|\, m, B) = w_U^D\, g_{\rm unr}(D) + w_R^D\, g_{\rm res}(D), \tag{13}$$

with $g_{\rm unr}(D)$ given by eq. 12 and $g_{\rm res}(D)$ modeled as a linear combination of log-normal distributions, we compute the probability that an object is unresolved, as well as the completeness and contamination analogously to eqs. 2-4.

3.5. *The Performance of the D Mixture Model*

Here we again use HST labels to evaluate completeness and contamination for the stellar multi-band D-based samples and compare them with the single-band performance. As before, we focus on the "transition" magnitude range, $24 \leq r < 25$, in two intervals, $24 \leq r < 24.5$ and $24.5 \leq r < 25$.

Figure 13 compares the empirical performance of the different band combinations in these two magnitude intervals.

APPENDIX

A toy model for D distributions

Figures 14 and 15 illustrate simulated unresolved and resolved populations and comparisons with model predictions.

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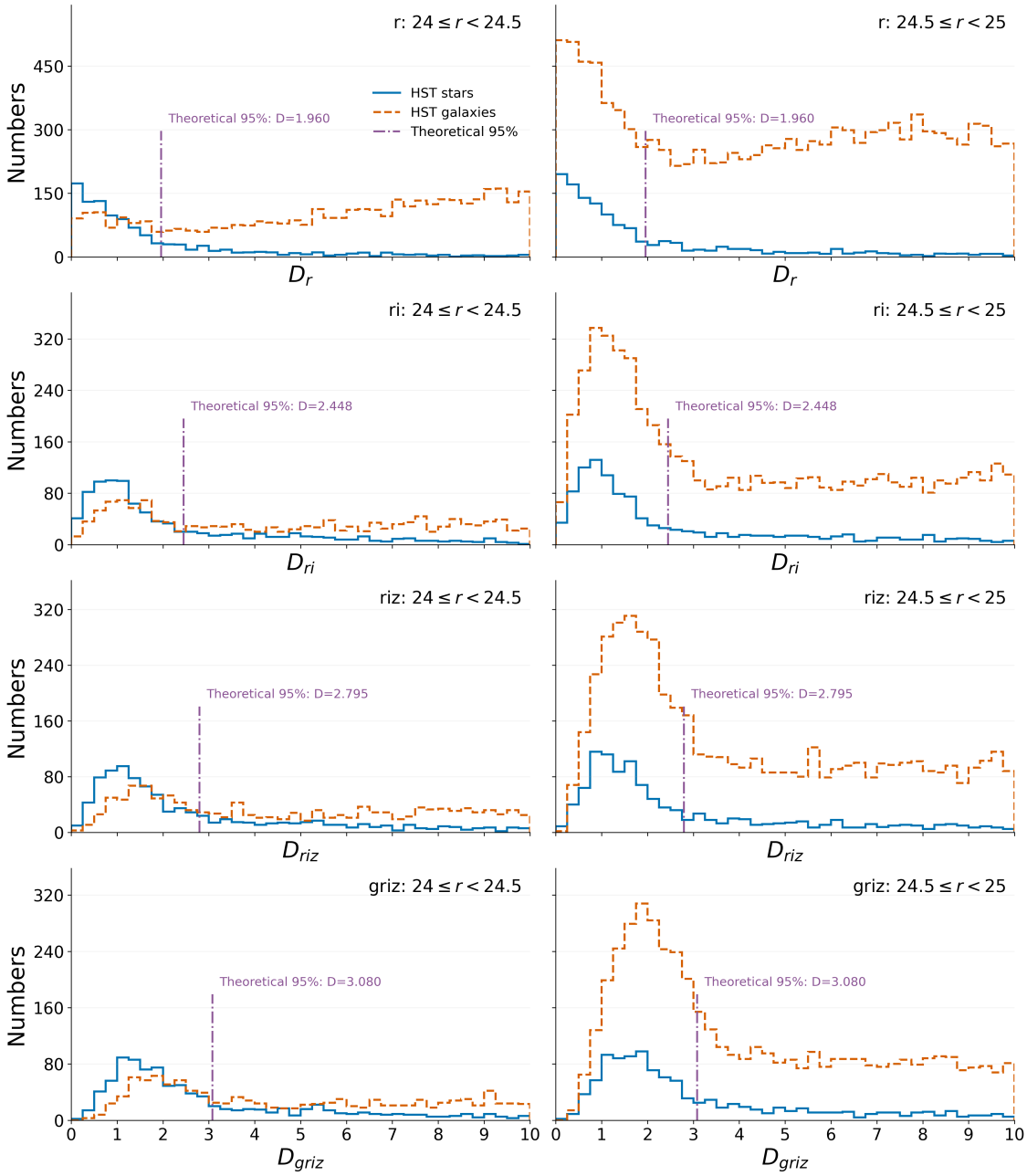


Figure 12. Observed D distributions of retained HST stars (blue) and HST galaxies (orange). Rows show r, ri, riz, and $griz$; columns show $24 \leq r < 24.5$ and $24.5 \leq r < 25$. Histograms show object counts in bins of width $\Delta D = 0.25$, over $0 \leq D \leq 10$. Half-height vertical lines mark the theoretical χ_N 95% thresholds: $D = 1.960$, 2.448, 2.795, and 3.080 for the four rows, respectively.

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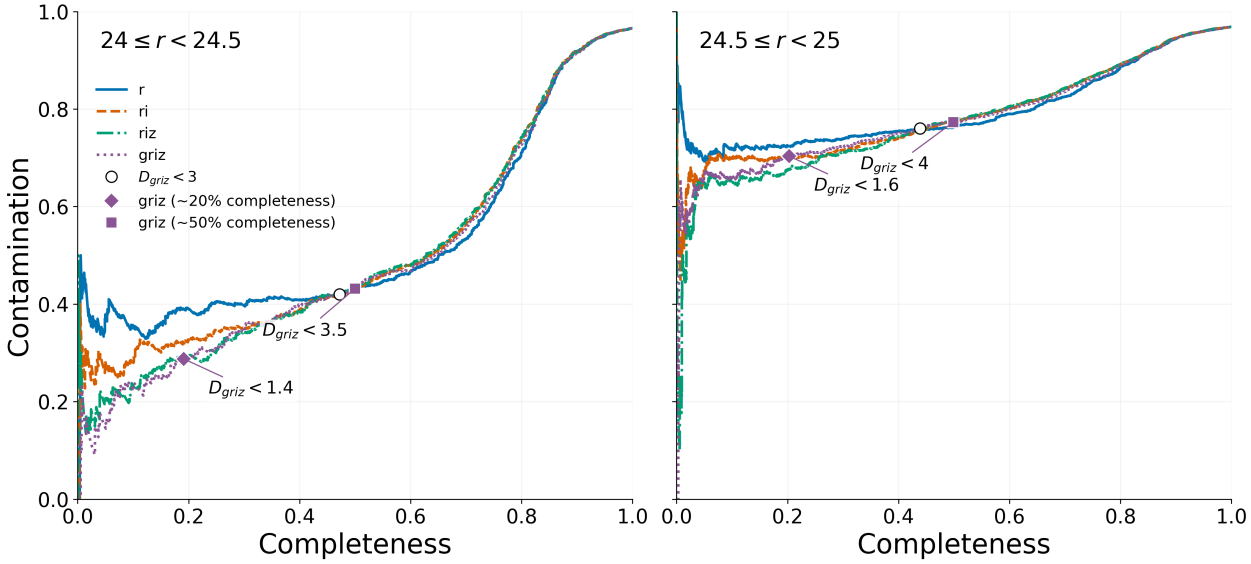


Figure 13. Empirical stellar completeness and galaxy contamination for strict $D_B < D_{\rm cut}$ selections using r, ri, riz, and $griz$. Left: $24 \leq r < 24.5$; right: $24.5 \leq r < 25$. All curves within a panel use the same retained-HST reference sample with valid D in all required bands. The open circles mark $D_{griz} < 3$. Diamonds and squares mark $griz$ thresholds nearest 20% and 50% measured stellar completeness on a 0.1-spaced threshold grid; the actual D thresholds are labeled. Objects beyond the Fig. 12 plotting window remain in the completeness and contamination denominators.

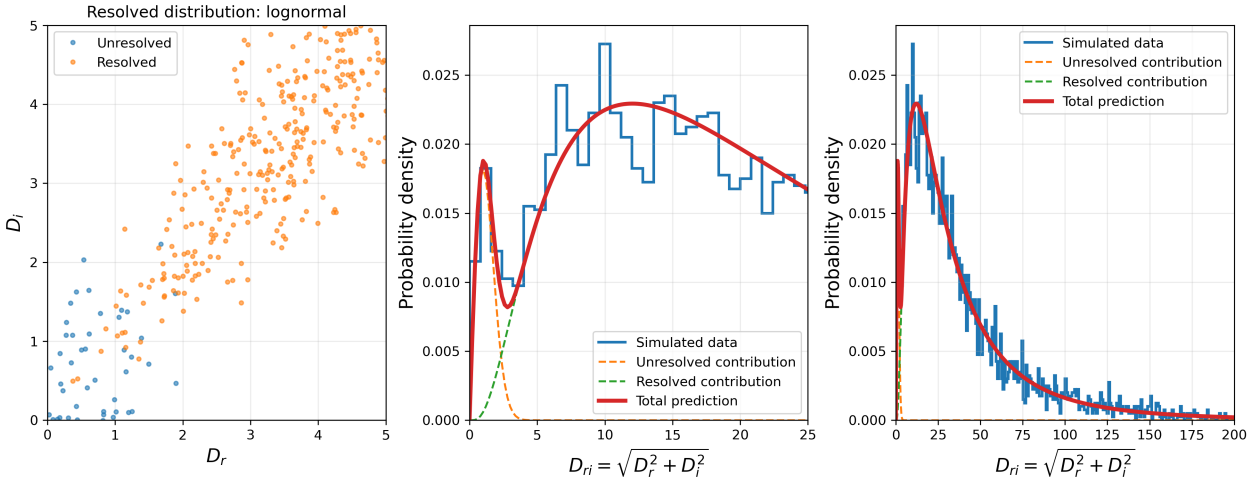


Figure 14. Illustrative simulated unresolved and resolved populations (left), with comparisons of simulated histograms, component predictions, and total predictions (middle and right).

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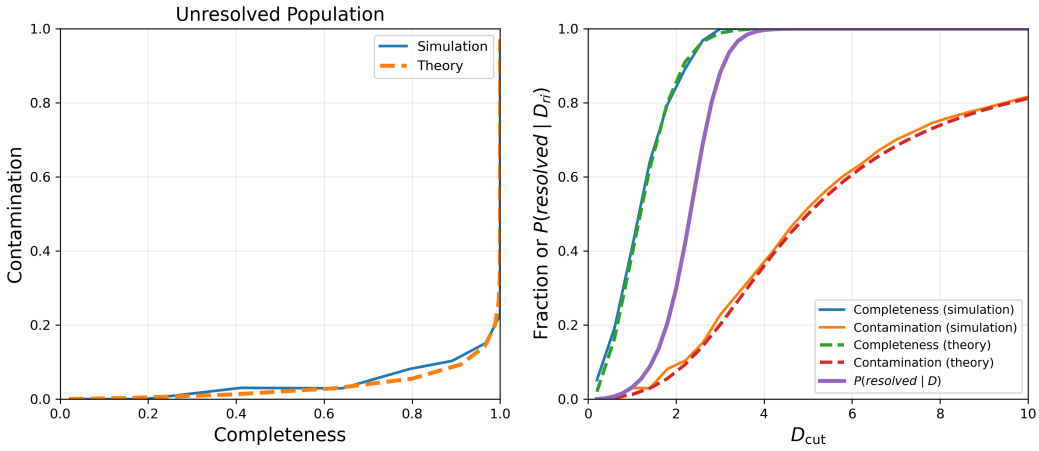


Figure 15. Simulation and theory comparisons for the illustrative model. Left: contamination versus completeness. Right: completeness and contamination versus $D_{\rm cut}$, together with the predicted probability of belonging to the resolved population.